Supplementary Material for "ContractKV: Certified Prompt Compilation for Transactional Prefix Reuse in Multi-Tenant LLM Services"

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I. Current-vLLM Relevance Audit

A. Protocol and Runtime Identity

This audit tests whether ContractKV's structural opportunity and integration objects remain present in a current runtime. It does not repeat the full service matrix or claim that the frozen vLLM 0.10.2 PSTP patch applies unchanged.

We use the official installed `vllm==0.25.1+cu129` wheel, Qwen3-8B in FP16, and one NVIDIA RTX 3090. Native automatic prefix caching is enabled. Five RULER `niah_single_1` cases and five LongBench HotpotQA cases are replayed under raw order and the same previously qualified frozen layout. Each cell runs in three fresh server processes, yielding 60 measured requests. Serving uses `max_model_len=20000`, `max_num_seqs=1`, and a 64-token output limit.

The optional FlashInfer sampler is disabled because the host's default CUDA toolkit predates the wheel runtime. Both layouts use this setting, all caches and logs reside on the data volume, and no vLLM source file is modified. The audited scheduler and cache-manager files match the installed wheel's `RECORD` hashes.

B. Real Serving Results

TABLE I
Native vLLM 0.25.1 APC performance. Token counts come from server metrics.

Workload	Layout	Mean TTFT	p95 TTFT	Hit tok.	New tok.
RULER	Raw	295.66	302.48	32	1,172
RULER	Compiled	**59.59**	**66.56**	1,136	**68**
HotpotQA	Raw	4,754.28	5,402.30	32	15,658.2
HotpotQA	Compiled	**153.60**	**174.79**	15,616	**74.2**

TABLE II
Paired behavior audit.

Workload	Layout	Task pass	Pair equal
RULER	Raw	100%	–
RULER	Compiled	100%	100%
HotpotQA	Raw	60%	–
HotpotQA	Compiled	60%	100%

The compiled layout reduces mean TTFT by 79.84% on RULER and 96.77% on HotpotQA. Observed new-prefill work falls by 94.20% and 99.53%, respectively. All requests complete. Every paired raw/compiled output is identical, and both the mean and minimum paired task-score differences are zero.

The absolute HotpotQA task pass rate is 60% in both cells. It is neither an official LongBench aggregate nor a ContractKV-induced regression. This small audit uses task scores only to test paired behavior preservation; the main paper reports the full quality protocols.

C. Source Portability Audit

TABLE III
vLLM 0.25.1 integration-object audit.

Requirement	Status
Layout admission	Available (`Scheduler.add_request`)
Prefix observation	Available (`get_computed_blocks`)
Slot allocation	Available (`allocate_slots`)
Cache publication	Available; interface adaptation required
Partial-block registration	Available; remains prefix-chain metadata
Frozen patch compatibility	Not drop-in

Native vLLM still searches a longest prefix chain rather than arbitrary non-contiguous stable segments. The admission, lookup, allocation, and publication object classes remain available, but their signatures and publication internals differ from the frozen implementation. The compiler's frozen output can therefore be consumed by the current runtime, whereas PSTP requires a version-specific backend adaptation.

This audit supports architectural relevance and adaptation feasibility, not full implementation equivalence across versions. Machine-readable request records, native metric summaries, source hashes, and the protocol manifest are available in the public artifact: https://github.com/zhouz4884-coder/contractkv.

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